🌸 Smriti — Research Dossier & Clinical Data Analysis

Comprehensive Evidence Base: Neuropathology, Epidemiological Data, NER Disparities & Open Source Landscape

LASI-DAD NATIONAL STUDY | LANCET PUBLIC HEALTH | WHO DEMENTIA REPORT | ARDSI GUWAHATI / MIZORAM | SIH26003 MDONER

1. Clinical Foundation: What Actually Happens in Dementia?

Dementia is not normal aging. It is a progressive neurodegenerative syndrome characterized by synaptic dysfunction, micro-vascular damage, and widespread neuronal death that impairs daily living activities (ADLs).

🧬 Pathophysiology & Subtypes

- **Alzheimer's Disease (AD) (~65%):** Extracellular Amyloid-β (Aβ) plaques + Intracellular hyperphosphorylated Tau neurofibrillary tangles lead to massive cholinergic neuron loss in the basal forebrain and hippocampal atrophy.
- **Vascular Dementia (VaD) (~20% — *highest in NER*):** Chronic cerebral hypoperfusion, micro-infarcts, and lacunar strokes driven by uncontrolled hypertension and arterial stiffness.
- **Frontotemporal & Lewy Body (FTD/LBD) (~15%):** Frontal/temporal atrophy causing early disinhibition and alpha-synuclein aggregates causing hallucinations.

🧠 5 Core Cognitive Domains Affected

- **1. Episodic & Working Memory:** Hippocampal failure; inability to recall recent meals or family names. *(Trained by Smriti's Market Day Basket)*
- **2. Executive Function:** Prefrontal cortex deficit; losing ability to brew tea or sequence clothes. *(Trained by Daily Routine Sequencer)*
- **3. Facial Agnosia:** Fusiform gyrus disconnect; alienating family. *(Trained by Faces & Family Recall)*
- **4. Auditory Attention:** Temporal lobe deficit. *(Trained by Sound & Rhythm Match)*
- **5. Semantic Discrimination:** Category confusion. *(Trained by Odd One Out)*

2. Raw Epidemiological Data & Demographic Statistics

Jurisdiction / Scope	Prevalence / Numbers	Source, Key Demographic Insights & Projections
Global Burden	**~55 Million** (2026) → **139M** by 2050	**WHO / Alzheimer's Disease International (ADI):** Over 60% of all dementia patients live in Low & Middle Income Countries (LMICs). Global cost exceeds $1.3 Trillion annually.
India National (60+)	**8.8 Million** (2026) **7.4% – 8.44%** prevalence	**LASI-DAD Study (Lancet Public Health / PLOS ONE):** Nationally representative study of seniors. Over 17.6% (24M+) suffer from Mild Cognitive Impairment (MCI). Projected to hit 14.3M by 2036.
Rural vs Urban Disparity	**68% Rural** vs 32% Urban	Rural India bears over two-thirds of the total burden due to limited awareness and specialist deficit.
Gender & Literacy	Females: **9.0%** (vs 5.8%) Non-literate: **10.1%**	Significantly higher prevalence in elderly women and seniors with zero formal schooling due to lower cognitive reserve.
North-Eastern Region (NER)	**~280,000 – 350,000** cases across 8 states	Assam accounts for ~180,000+ cases. Mizoram and Meghalaya show the fastest-growing vascular risk trajectories.

3. Why Does the North-Eastern Region (NER) Suffer More Than Rest of India?

1. The Vascular Risk Triad (Hypertension + Diet)

- **Uncontrolled Hypertension:** Studies in Assam (Kamrup ~33%) and Mizoram show hypertension rates 15–20% higher than the national average, directly triggering **Vascular Dementia**.
- **High-Sodium Dietary Practices:** Heavy consumption of alkali water (*Khar*), fermented salted fish (*Ngari*, *Shidol*), smoked meats, and betel nut (*Tamul*) damages micro-vasculature.

2. The "Neurology Desert" (Extreme Specialist Deficit)

- **85% Specialist Concentration:** Over 85% of all practicing neurologists in the entire 8 states are located in Guwahati (GMCH, GNRC, Apollo).
- **Zero Rural Neurologists:** Hill states like Arunachal, Nagaland, and Mizoram have near-zero full-time cognitive neurologists outside capital cities, requiring 12-hour mountain journeys for basic CT/MRI scans.

3. Topography & Connectivity Chasm

- Rugged mountainous terrain prevents routine outpatient clinical visits for frail, mobility-impaired elderly.
- Intermittent rural 2G/3G networks cause conventional cloud-only health apps to crash and lose vital adherence telemetry.

4. Linguistic Diversity & Cultural Alienation

- NER houses **over 220+ indigenous dialects** (Assamese, Bodo, Khasi, Garo, Mizo, Meitei, Nagamese).
- Standard Western cognitive tests (MMSE) use foreign concepts (e.g. US seasons), yielding false-positive decline scores.
- **Stigma:** 90%+ cases written off as normal aging (*"বুঢ়া বয়সৰ পাহৰণি"*), leading to diagnosis only at late Stage 3/4.

4. Major NGOs, Clinical Bodies & Government Initiatives in India

Organization / Body	HQ & Operational Scope	Key Clinical Programs & Regional NER Presence
ARDSI (Alzheimer's & Related Disorders Society of India)	Kochi / National Apex NGO (Est. 1992)	Publishes the authoritative *Dementia in India Report*, runs daycare centers, caregiver training. Active **ARDSI Guwahati Chapter** (Dr. H. K. Goswami) & **Mizoram Chapter (Aizawl)**.
NIMHANS Bangalore	National Institute of Mental Health & Neurosciences	Developed ICMR-NCDC Cognitive Assessment tools; operates national 24×7 **Tele-MANAS (14416)** tele-counseling helpline across all NER states.
Dementia India Alliance (DIA)	National Non-Profit Network	Operates "DemClinic" online clinical support, caregiver mutual-aid networks, and legal advocacy for elder rights.
Nightingales Medical Trust (NMT)	Bangalore / Pan-India	Specialized dementia daycare centers, memory screening mobile vans, and caregiver respite training modules.
SCARF India (DEMCARES)	Chennai (WHO Collab. Centre)	Community-based dementia screening toolkits designed for rural ASHA and Anganwadi healthcare workers.
Ministry of DoNER & NEC	Govt. of India / Shillong	Sponsors regional telemedicine infrastructure, rural health nodes, and SIH26003 innovation challenges.

5. Existing Market Apps vs. Why They Fail in India & NER

App / Platform	Target Demographic	Critical Vulnerabilities / Why They Fail in North-Eastern India
CogniFit / Lumosity / BrainHQ	US / European Commercial ($15–$20/month)	• Expensive monthly subscription model unaffordable for rural families. • English-only abstract geometric puzzles with zero cultural resonance. • Requires high-speed broadband; zero offline background sync.
MindMate / Timeless / CareZone	Western Dementia Patients & NHS/Medicare	• Geared around Western healthcare pathways; lacks Indian joint-family context. • No voice STT/TTS in Assamese or Hindi; no WhatsApp reminder outreach. • Crashes on slow 2G rural hill connections.
🌸 **SMRITI (Our Platform)**	**North-Eastern & Indian Elderly Demographics**	✓ **100% Free & Open-Access:** Accessible to every rural clinic. ✓ **Culturally Attuned:** Kaji Nemu, Bihu Dhol, Tulsi puja, Assam Tea. ✓ **100% Offline-First PWA:** Workbox + IndexedDB + Biometric WebAuthn. ✓ **Multimodal Voice & WhatsApp:** EN/HI live voice, Assamese UI, wa.me bot. ✓ **Clinical ML Telemetry:** Live adaptive difficulty & Red Flag caregiver alerts.

6. Verified Open-Source GitHub Repositories & Research Papers

Repository / Paper Title	Repository URL / DOI Reference	Core Methodology & Academic Relevance
LASI-DAD Diagnostic Study (Lancet Public Health 2023)	doi.org/10.1016/S2468-2667(22)00301-7 Portal: https://lasi-dad.org	Nationally representative clinical dataset of 4,000+ Indian seniors validating 7.4% national dementia prevalence.
NEUROHACK2022_Dementia (Clinical Dementia Rating Model)	github.com/shreyasgite/dementianet github.com/DEMON-NEUROHACK	Machine learning pipeline using PCA + SVM and Random Forest to predict Clinical Dementia Rating (CDR) scores.
DementiaVoiceAnalyzer (Speech Acoustic Biomarkers)	github.com/Butovens/DementiaVoiceAnalyzer	Extracts acoustic speech features (pitch, shimmer, jitter via openSMILE) with SVM/LSTM models to detect early cognitive decline.
Alzheimer-Detection (NLP) (Spontaneous Speech Analysis)	github.com/42bismuth/Alzheimer-Detection	Full-stack React + Python NLP pipeline analyzing hesitation pauses, lexical richness, and speech-to-text patterns.
Cognitive Stimulation Therapy (NICE Clinical Guidelines CG42)	nice.org.uk/guidance/cg42 World Alzheimer Report (ADI)	Gold standard non-pharmacological evidence proving group CST improves cognitive scores by 6–9 months.